

Generative Diffusion and Sampling

Algorithms and Theory in High Dimensions

Yifan Chen

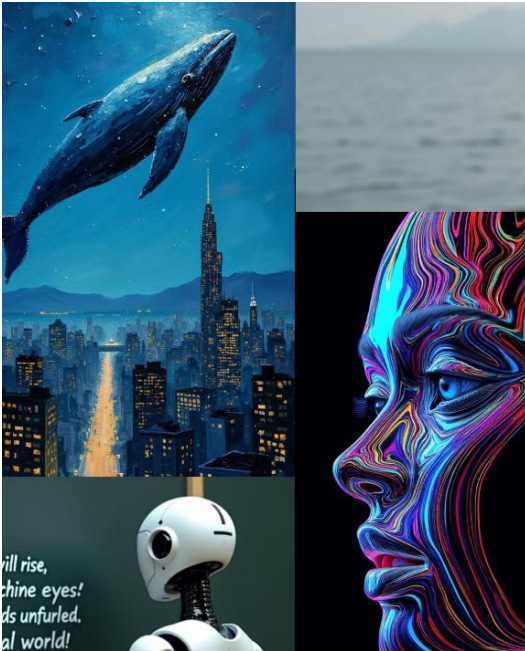
UCLA Mathematics

2026

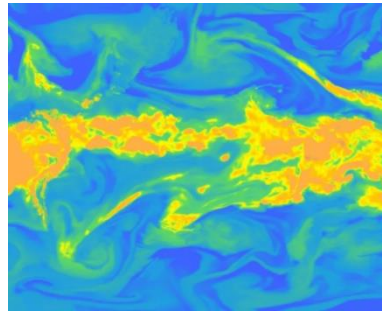
Generative Diffusions and Sampling

Sampling from probability distributions is a fundamental challenge in scientific computing, statistics, data science, and generative AI

Image



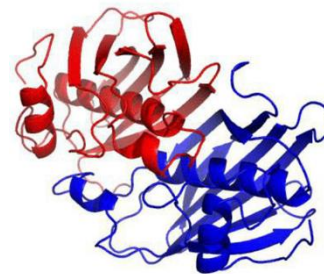
Weather



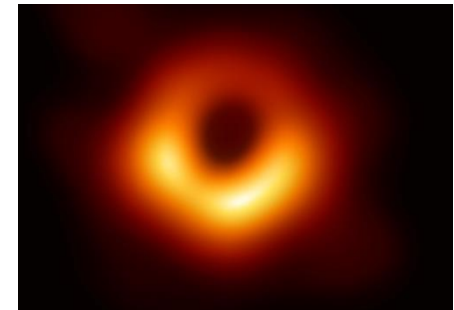
Material science



Molecular dynamics



Astrophysics



Physical models and data often exhibit complex structures with natural *probabilistic interpretations*

These probability distributions are often very *high dimensional*

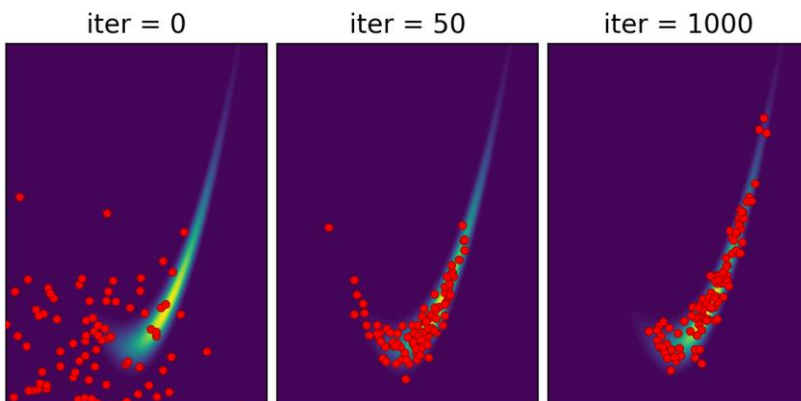
Mathematical Formulation

Problem setting

Goal: draw new samples from a density $\pi \propto \exp(-V)$ either through

1. queries to the potential V (*Monte Carlo sampling*)
2. given sampled data $\{x^{(i)}\}_{i=1}^N \sim \pi$ (*generative modeling*)

State-of-the-art methodology relies on building **dynamics of measures**



1. *Stochastic dynamics in MCMC*



2. *From noise to generating new data*

Mathematical Formulation

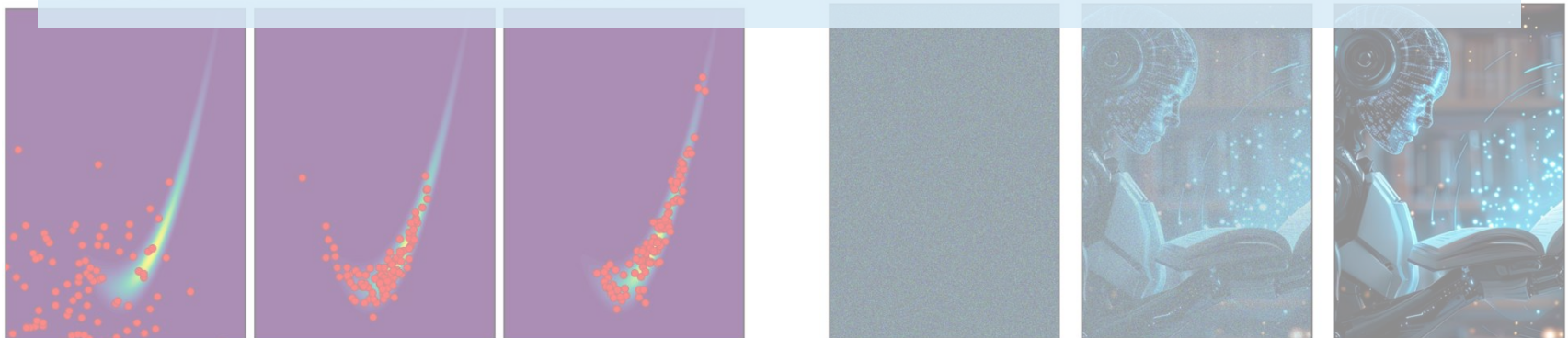
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Guiding research questions:

- State-of-the-art methodology relies on building *dynamics of measures*
- **Methods and design principles** for scientific applications
 - Rationale for their success in **high dimensions**



1. Stochastic dynamics in MCMC

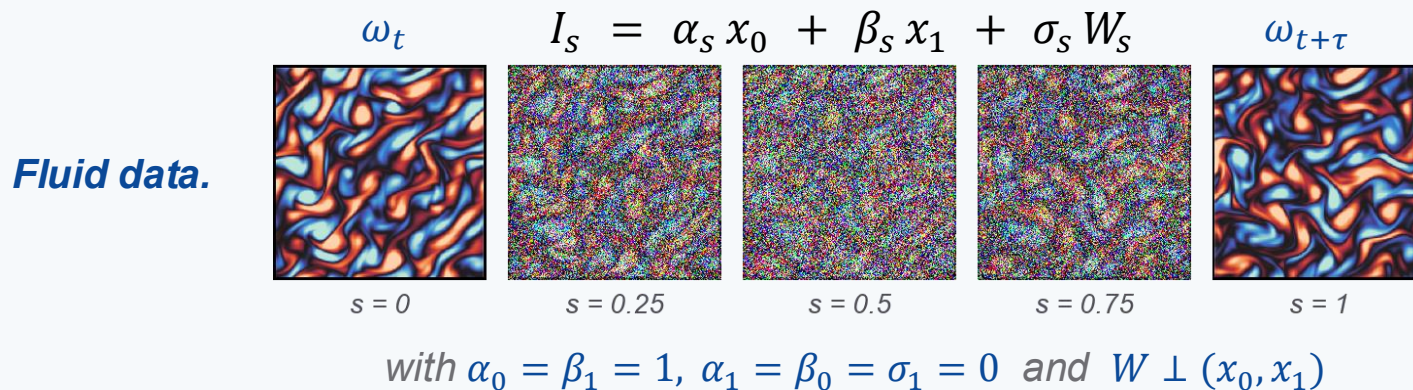
2. From noise to generating new data

Generative diffusion from a point source

[YC, Goldstein, Hua, Albergo, Boffi, Vanden-Eijnden 2024]

Setting: probabilistic forecasting — given current state $x_0 = \omega_t$, sample plausible futures $x_1 = \omega_{t+\tau}$ from the conditional $\rho^*(\cdot | x_0)$ where $(x_0, x_1) \sim \rho^*(x_0, x_1)$

Method: Diffusive interpolation based on collected data pairs $(x_0, x_1) = (\omega_t, \omega_{t+\tau})$



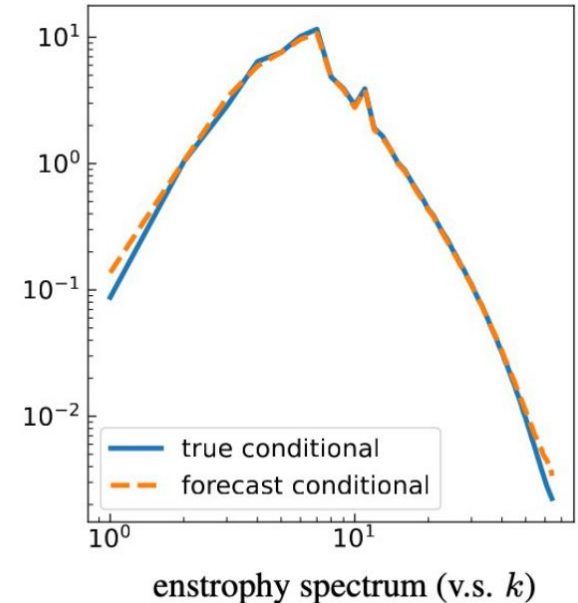
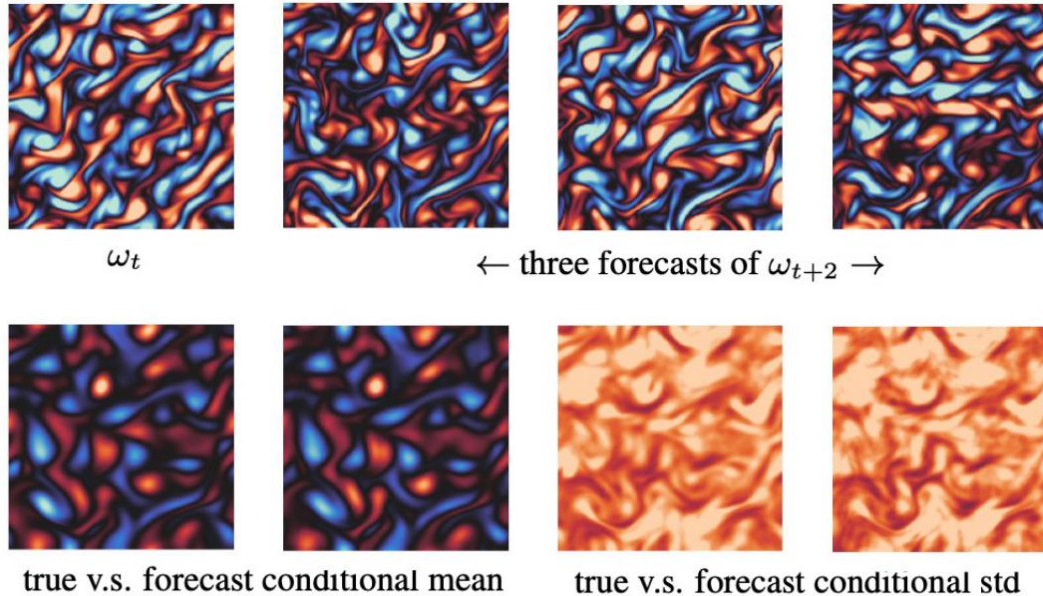
Theorem. With $b_s(x, x_0) = \mathbb{E}[\dot{\alpha}_s x_0 + \dot{\beta}_s x_1 + \dot{\sigma}_s W_s | I_s = x, x_0]$, $g_s = \sigma_s$

the SDE $dX_s = b_s(X_s, x_0) ds + g_s dW_s$, $X_0 = x_0$ forecasts $X_1 \sim \rho^*(\cdot | x_0)$

Training loss:
$$L_b[\hat{b}_s] = \int_0^1 \mathbb{E} [|\hat{b}_s(I_s, x_0) - \dot{\alpha}_s x_0 - \dot{\beta}_s x_1 - \dot{\sigma}_s W_s|^2] ds$$

Forecasting results — 2D stochastically forced NSE

Generation reproduces accurate conditional statistics.



lag $\tau = 2$ (autocorrelation $\sim 10\%$) · resolution 128×128 · 20K data pairs · 2M-parameter U-Net for the drift

Same methodology now adapted in large-scale weather forecasting at NVIDIA.

[Kossaifi et al., NVIDIA 2026]

Mathematical design questions: how to optimally choose the generative dynamics?

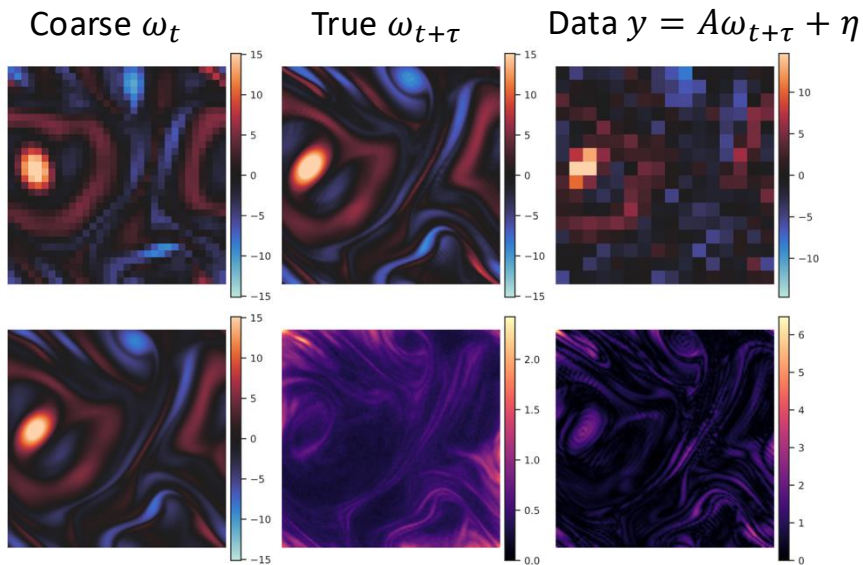
→ variational, multiscale structures and statistical / numerical principles

[YC, Vanden-Eijnden, Xu 2025] · [YC, Vanden-Eijnden 2025] · [YC et al. 2024]

Posterior sampling applications

Many scientific tasks formulated as posterior sampling:
combine a **generative prior** with **likelihood** $\propto \exp(-V)$

Bayesian data assimilation

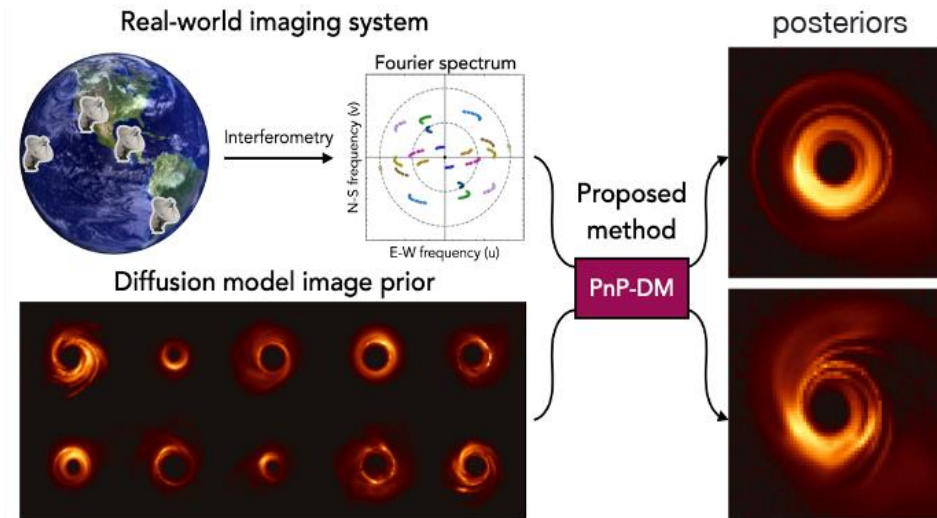


Forecast ensemble mean of $\omega_{t+\tau}$ Error of mean and ensemble std

Coarse ω_t + observation $y = A\omega_{t+\tau} + \eta$
Generative prediction + ensemble Kalman filter

[YC, Vanden-Eijnden 2026]

Black-hole imaging



Diffusion priors with interferometric likelihoods

Plug-and-play diffusion + Monte Carlo sampler
addresses challenging inverse problems

[Sun, Wu, YC, Feng, Bouman 2023]
[Wu, Sun, YC, Zhang, Yue, Bouman 2024]

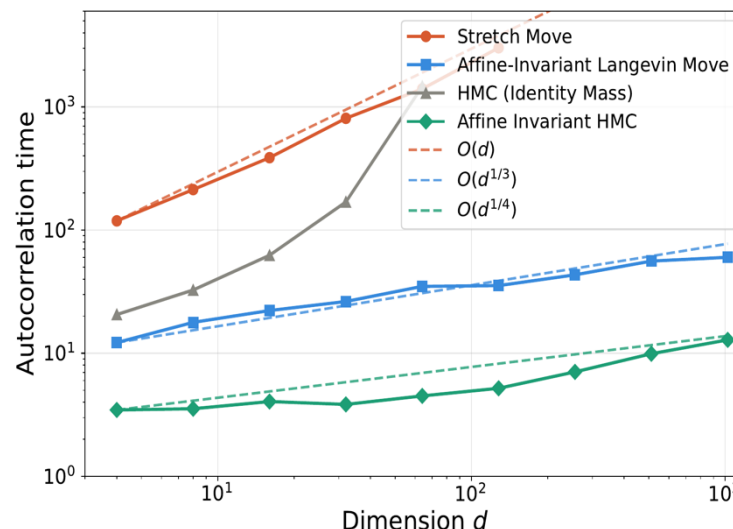
Improving high-dimensional scaling of sampling

Sampling from posterior $\pi \propto \exp(-V)$ in very high dimension d
— how to push the boundary for practical sampler tools?

- ▶ Affine invariant ensemble sampler (stretch move) in `emcee` package routinely used for Bayesian calibration in astrophysics (15k citations from 2012)
- ▶ They are reported to suffer dimensions $d > 20$
- ▶ We are working with Erik Petigura in astrophysics to develop new generation high dimensional sampler package for astro-posteriors

Thanks to DataX pilot program for the support!

Preliminary results



Theoretical results for high dimensional complexity of samplers

Theorem (informal): new gradient based affine invariant ensemble samplers based on Hamiltonian dynamics complexity scale better $O(d^{\frac{1}{4}})$ with dimension than `emcee` $O(d)$

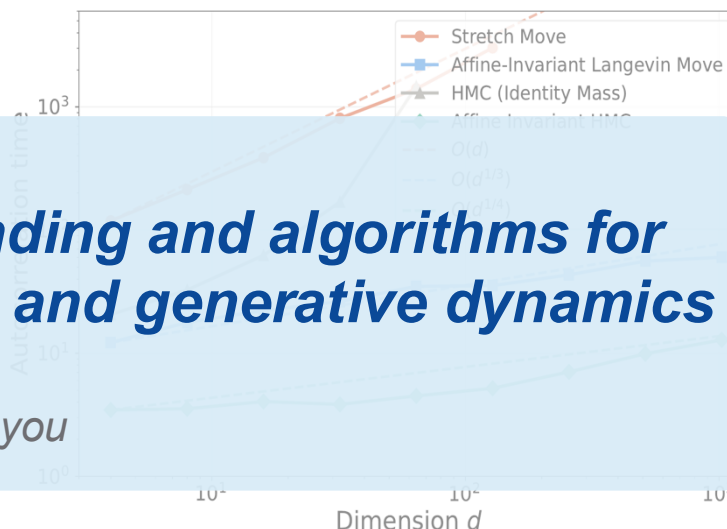
Theorem (informal): unadjusted sampler can have *dimension-free* complexity for low-dim observables

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Preliminary results



Towards better understanding and algorithms for high-dimensional sampling and generative dynamics

thank you

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